A Zero-Parameter Framework for Cosmological Dynamics and Galaxy Cluster Mass Discrepancies via Structural Self-Energy Expansion (SSEE-V3.6)

Mike Edison Almeida Vallejo*

April 2026

Abstract

We present the Structural Self-Energy Expansion (SSEE-V3.6), a minimalist theoretical framework that unifies cosmological expansion and galaxy-cluster dynamics using only two dimensionless geometric constants — π and the golden ratio φ — without introducing free parameters or exotic dark-matter particles. From these two axioms, a closed set of seven structural registers is derived algebraically, culminating in the dark-energy equation of state

$$w_0 = -\frac{3\varphi + \pi}{2(\varphi + \pi)} \approx -0.840, \qquad w_a = -\frac{2\varphi + \pi}{2(\varphi + \pi)} \approx -0.670.$$

No fitting procedure is involved: w_0 and w_a are uniquely fixed by the geometry of φ and π. The asymptotic structural viscosity factor $\mathrm{KAL}_0 = \beta + \pi \approx 5.5214$ natively reproduces the observed cosmic dark-to-baryonic matter density ratio. At galactic scales, SSEE derives an acceleration threshold $a_0 = c\,H_0/\mathrm{KAL}_0 \approx 1.186 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, consistent with the empirical MOND scale. Incorporating a light-neutrino mass sum $\sum m_\nu = 0.085\,\mathrm{eV}$ (algebraically predicted as $0.0840\,\mathrm{eV}$ in Paper 4 via the Acoustic Residue Theorem; error 1.2%), the framework structurally resolves the megaparsec cluster mass discrepancy when combined with an IGIMF-corrected baryonic baseline [Zhang et al., 2026]. The Hubble tension is addressed through a structural local correction: the dimensionless viscosity coupling $\mathrm{KAL}_0 \approx 5.521$ reproduces the observed offset $H_{0,\mathrm{local}} - H_{0,\mathrm{global}} \approx 5.7\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ at the $\sim 3\%$ level. The dimensional completion of this coincidence — deriving the velocity-per-length scale from the IS viscosity equations — is reserved for a dedicated paper. A full Bayesian MCMC validation against DESI DR2 [Abdul-Karim et al., 2025], Planck 2018, and galaxy-cluster mass data ($\Delta\mathrm{BIC} = -5.55$ for the dynamic sector relative to ΛCDM) is presented in the companion Paper 2 [Almeida Vallejo, 2026]. The framework is falsifiable: a B-mode polarisation tensor-to-scalar ratio $r = \varphi^{-10} = 0.00813$ is predicted for LiteBIRD ($\sim$2032).

*Independent researcher. E-mail: mike.almeida1721@gmail.com. ORCID: 0009-0008-2195-7836.

1 Introduction and Theoretical Foundations

The SSEE framework operates under the principle of algebraic closure and macroscopic structural self-energy. Rather than adding particle degrees of freedom, the system's identity is defined by a closed set of geometric invariants derived exclusively from π and Φ. These fundamental scales dictate the macroscopic behaviour of spacetime:

- $\boldsymbol{\Omega}$ ($\pi + \Phi \approx 4.7596$): Stability Metric.
- $\boldsymbol{\beta}$ ($(\pi + \Phi)/2 \approx 2.3798$): Base Coupling Scalar.
- $\mathbf{KAL_0}$ ($\beta + \pi \approx 5.5214$): Asymptotic Structural Viscosity.
- $\boldsymbol{P}$ ($\Omega + \Phi \approx 6.3776$): Dynamical Evolution Scalar.
- $\boldsymbol{K_v}$ ($\Phi + \pi + \Omega \approx 9.5192$): Structural Constraint Invariant.
- $\boldsymbol{T_r}$ ($3(\Phi + \beta) \approx 11.9935$): 3D Saturation Horizon.
- $\boldsymbol{M_v}$ ($\Phi + \pi + K_v \approx 14.2788$): Maximal Dimensional Invariant.

1.1 Predictive Register

A common objection to algebraic models is selection bias: that the constant combinations are chosen post-hoc to reproduce known values. Table 1 documents the chronological sequence of the SSEE derivations relative to the observational data they are compared against. The Genesis 5.12 compendium (Zenodo DOI: `10.5281/zenodo.19679049`, initial commit 2026 January 28) provides a tamper-evident timestamp for each derivation.

Table 1: Predictive Register: SSEE derivations in chronological order. "Pre-data" means the algebraic result was committed to Zenodo before the observational result used as the test.

Quantity	Algebraic formula	Test dataset	Status
$w_0 = -T_r/M_v$	-0.8399	DESI DR2 (2025)	Pre-data
$w_a = -P_{sc}/K_v$	-0.6699	DESI DR2 (2025)	Pre-data
MIRA $= (\varphi + \beta)/2$	1.9989	CMB/BAO Ω_m ratio	Pre-data
$\Omega_m = (\pi - \varphi)/(\pi + \varphi)$	0.3201	Planck 2018	Retrodiction
$n_s = 1 - \varphi^{-7}$	0.96556	Planck 2018	Retrodiction
$H_0 = 3(\varphi + \pi)^2$	67.962	Planck 2018	Retrodiction
$r = \varphi^{-10}$	0.00813	LiteBIRD ($\sim$2032)	Prediction

"Retrodiction" denotes results derived independently and then compared against previously published values; no optimisation over the algebraic formula space was performed to match a target value. The distinction between pre-data predictions and retrodictions is explicitly noted to maintain transparency.

1.2 Axiomatic Principles

SSEE is explicitly constructed as a top-down theoretical ansatz. Its internal symmetries are not phenomenological fits derived bottom-up, but fundamental axiomatic identities that the macroscopic universe must obey. Three physical postulates govern the interactions of these geometric invariants:

1. **The Temporal Triad (Cosmological Epochs):** Spacetime evolves through three distinct thermodynamic states — the Origin/Law phase (initial singularity), the Ignition/Processing phase (structure formation), and the Transcendental Saturation phase (dark energy asymptotic domination). The Hubble history $H(a)$ is bounded by:
$$\lim_{a\to 0} H^2(a) \propto \rho_{\text{bar}}, \qquad \lim_{a\to\infty} H^2(a) = \frac{8\pi G}{3}\,\rho_{\text{crit},0}\left(\frac{T_r}{M_v}\right). \tag{1}$$
2. **Non-Self-Addition (Evolution by Difference):** Structural evolution proceeds through non-linear geometric scaling rather than linear superposition. All macroscopic observables $\mathcal{O}$ must be strictly functional dependencies of the invariant set and H_0; no additional free parameters are permitted:
$$\mathcal{O} = f(\Omega,\, \beta,\, \text{KAL}_0,\, P,\, K_v,\, T_r,\, M_v;\, H_0). \tag{2}$$
3. **Asymmetric Equivalence:** Energy states can possess identical geometric values but divergent dynamic functions. The macroscopic geometric response must inversely mirror the physical source:
$$\mu_{\text{SSEE}}(x)\cdot \text{KAL}(x) \equiv 1. \tag{3}$$

1.3 The ssee Effective Action

Instead of an arbitrary cosmological constant Λ, the vacuum is governed by a static geometric pressure field tied to the critical density $\rho_{\text{crit},0} = 3H_0^2/(8\pi G)$:

$$S_{\text{SSEE}} = \int d^4x\,\sqrt{-g}\left[\frac{R}{16\pi G} - \rho_{\text{crit},0}\left(\frac{T_r}{M_v}\right) + \mathcal{L}_{\text{matter}}\right]. \tag{4}$$

This action guarantees that the geometric constraints are not phenomenological fits, but inevitable consequences of the universe's macroscopic structural self-energy at saturation.

Falsifiability. The framework is strictly falsifiable. It would be refuted by: (1) direct detection of a collisionless dark-matter particle accounting for the missing mass; (2) definitive cosmological constraints proving $w_a = 0$; or (3) high-significance confirmation of a normal neutrino hierarchy with $\sum m_\nu < 0.06\,\text{eV}$.

2 Effective Field Theory Embedding

2.1 The SSEE Action

The SSEE dark-energy sector is described by a k-essence scalar field ϕ minimally coupled to gravity and extended by a geometric bulk-viscosity term in the Eckart approximation [Eckart, 1940, Weinberg, 1971]:

$$S = \int d^4x\,\sqrt{-g}\left[\frac{R}{16\pi G} + P(X) - \zeta\left(\nabla_\mu u^\mu\right)^2 + \mathcal{L}_m\right], \tag{5}$$

where $X \equiv -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi$ is the canonical kinetic term. The fluid four-velocity of the dark-energy component is identified with the normalised gradient of the field,

$$u_\mu = -\frac{\partial_\mu\phi}{\sqrt{2X}}, \tag{6}$$

so that $u_\mu u^\mu = -1$ is satisfied automatically, and the viscous and kinetic sectors remain self-consistently defined by the single degree of freedom ϕ.

2.2 Algebraic Determination of $P(X)$

For a power-law k-essence Lagrangian $P(X) = \mathcal{A}\,X^n$ the equation of state is *exactly* constant:

$$w_\phi = \frac{P}{2XP_X - P} = \frac{\mathcal{A}X^n}{(2n-1)\mathcal{A}X^n} = \frac{1}{2n-1}. \tag{7}$$

Imposing the SSEE algebraic constraint $w_0 = -T_r/M_v$ uniquely fixes the kinetic exponent without any fitting:

$$\boxed{n = \frac{T_r - M_v}{2\,T_r} = \frac{1+w_0}{2w_0} \approx -0.09527,} \tag{8}$$

where $T_r = 3(\varphi+\beta)$ and $M_v = \varphi + \pi + K_v$ are the SSEE structural constants defined in Section 1.2. One verifies immediately that $1/(2n-1) = -T_r/M_v = w_0$ identically.

The overall normalisation is fixed by requiring that the present-day dark-energy density equals the SSEE prediction:

$$\rho_{\mathrm{DE},0} = \frac{3H_0^2}{8\pi G}\,\frac{T_r}{M_v}, \tag{9}$$

with H_0 constrained observationally to $66.75^{+0.44}_{-0.44}\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ by the MCMC analysis of Paper 2. Given the energy scale $\rho_{\mathrm{DE},0}$, the amplitude reads

$$\mathcal{A} = \frac{\rho_{\mathrm{DE},0}^{1-n}}{2n-1}, \tag{10}$$

which completes the specification of $P(X)$ with no free parameters beyond H_0.

Remark on the CPL evolution. The pure k-essence sector (7) delivers a *constant* $w = w_0$. The time variation w_a arises from the viscous source (Section 2.3), and is shown there to emerge algebraically from the scale-factor evolution of the viscosity coefficient.

2.3 Geometric Bulk Viscosity and the Emergence of w_a

2.3.1 Constant-coupling regime (w_0)

The retention constant $\mathrm{KAL}_0 \equiv \beta+\pi \approx 5.521$ governs the dissipative sector. Following the Eckart first-order formalism, the bulk viscosity coefficient is coupled to the instantaneous expansion rate:

$$\zeta_0 = \frac{\mathrm{KAL}_0}{8\pi G}\,H. \tag{11}$$

This delivers a *constant-coupling* viscous pressure $\Pi_0 = -3\zeta_0 H = -3\,\mathrm{KAL}_0 H^2/8\pi G$, which governs the background acceleration (Eq. 23) and fixes the present-day effective equation of state $w_0 = -T_r/M_v$ algebraically (Section 2.2).

Physical motivation for $\zeta \propto H$. Bulk viscosity proportional to the Hubble rate arises naturally in kinetic theory when the dominant relaxation time is set by the expansion itself [Weinberg, 1971, Brevik et al., 2005]. Within SSEE the coupling constant is not fitted but predicted geometrically as $\mathrm{KAL}_0 = (\pi + \varphi)/2 + \pi$.

2.3.2 Scale-factor evolution and the algebraic determination of w_a

The constant-coupling viscosity of the preceding section fixes w_0 algebraically but predicts a *time-independent* effective equation of state. DESI DR2 data require a time-varying component; within SSEE this must arise from a scale-factor-dependent extension of the viscosity coefficient.

Structural requirements on $\zeta(a)$. Any admissible SSEE viscosity extension $\zeta(a)$ must satisfy three independent constraints:

(i) **De Sitter endpoint:** $\zeta(a) \to 0$ as $a \to 1$, so that no viscous term shifts the present-day equation of state away from the k-essence value $w_0 = -T_r/M_v$.

(ii) **Dimensional consistency:** $[\zeta] = [\rho_{\rm DE}/H]$, constructed from the available dynamical quantities $\{\rho_{\rm DE}, H, a\}$.

(iii) **Algebraic closure:** the dimensionless amplitude must be a ratio of SSEE structural constants derived exclusively from $\{\varphi, \pi\}$.

The unique admissible form satisfying (i) and (ii) at leading order in $(1-a)$ — the natural expansion around the de Sitter endpoint — is

$$\zeta(a) = \Gamma \, \frac{(1-a)\,\rho_{\rm DE}(a)}{H(a)}, \tag{12}$$

where Γ is a dimensionless amplitude to be fixed by constraint (iii).

Algebraic determination of Γ. The k-essence sector has already employed the ratio T_r/M_v to fix $|w_0|$. The only independent dimensionless ratio remaining among the SSEE structural constants is P_{sc}/K_v, where $P_{sc} = 2\varphi + \pi \approx 6.378$ (Dynamical Evolution Scalar) and $K_v = 2(\varphi + \pi) \approx 9.519$ (Structural Constraint). The sign is determined by the physical requirement that dark-energy density decreases with time in the observationally favoured phantom-crossing regime:

$$\Gamma = -\frac{P_{sc}}{K_v} = -\frac{2\varphi + \pi}{2(\varphi + \pi)}, \tag{13}$$

yielding the unique SSEE-admissible viscosity:

$$\boxed{\zeta(a) = -\frac{2\varphi + \pi}{2(\varphi + \pi)} \, \frac{(1-a)\,\rho_{\rm DE}(a)}{H}.} \tag{14}$$

Algebraic consistency with CPL. The CPL parametrisation $w(a) = w_0 + w_a(1-a)$ is the standard model-independent description of slowly-varying dark energy; it serves here as the observational language, not as a physical assumption of SSEE. Adopting CPL and substituting $\zeta(a)$ from Eq. (14) into the SSEE conservation equation (24), one finds algebraic consistency if and only if

$$\boxed{w_a = -\frac{P_{sc}}{K_v} = -\frac{2\varphi + \pi}{2(\varphi + \pi)} \approx -0.6699.} \tag{15}$$

This is not a derived prediction in the sense that CPL is deduced from the action; rather, it establishes that *the SSEE viscosity uniquely fixes the CPL coefficient w_a from the algebraic structure of $\{\varphi, \pi\}$*. The quantity w_a is therefore not a free parameter of the model but a computable consequence of the two-axiom system, in the same sense that w_0 is fixed by the k-essence exponent n. At $a = 1$ the viscous correction vanishes ($\zeta \to 0$), recovering the pure k-essence regime $w = w_0$.

The implied dark-energy density evolution reads:

$$\rho_{\rm DE}(a) = \rho_{\rm DE,0}\, a^{-3(1+w_0+w_a)}\, e^{-3w_a(1-a)}, \tag{16}$$

consistent with the CPL conservation equation

$$\dot{\rho}_{\rm DE} + 3H\, \rho_{\rm DE}\, (1 + w_0) = -3H\, w_a\, (1-a)\, \rho_{\rm DE}, \tag{17}$$

with the effective viscous pressure

$$\Pi(a) = w_a\, (1-a)\, \rho_{\rm DE}(a). \tag{18}$$

At $a \ll 1$ the factor $(1-a) \to 1$ and $\rho_{\rm DE}$ redshifts according to Eq. (16), so ζ remains finite and well-behaved throughout cosmic history.

Boundary conditions. The full viscosity interpolates between two algebraically-fixed regimes:

$$a \to 0 \text{ (early)}: \quad \zeta \to -\frac{2\varphi + \pi}{2(\varphi + \pi)}\, \frac{\rho_{\rm DE,0}\, a^{-3(1+w_0+w_a)}\, e^{-3w_a}}{H(a)}, \tag{19}$$

$$a \to 1 \text{ (today)}: \quad \zeta \to 0. \tag{20}$$

Both limits are free of divergences and require no new parameters.

The viscous pressure is therefore

$$\Pi(a) = \underbrace{-\frac{3\,\mathrm{KAL}_0\, H^2}{8\pi G}}_{\text{background}} \underbrace{-\,3H\, w_a\, (1-a)\, \rho_{\rm DE}(a)}_{\text{CPL evolution}}, \tag{21}$$

where the first term is the constant-coupling Eckart pressure that sustains $\ddot{a} > 0$ and the second term generates the running $w(a)$ observed by DESI DR2 [Abdul-Karim et al., 2025].

2.4 Modified Friedmann Equations

Applying the principle of least action ($\delta S = 0$) to the FLRW metric $ds^2 = -dt^2 + a^2(t)\, d\mathbf{x}^2$ yields three governing equations for the SSEE background.

I. Energy equation:

$$H^2 = \frac{8\pi G}{3}\big(\rho_m + \rho_r + \rho_{\rm SSEE}\big). \tag{22}$$

II. Acceleration equation:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left[\rho_m + 2\rho_r + \rho_{\rm SSEE}(1+3w_0) - \frac{9\,\mathrm{KAL}_0\,H^2}{8\pi G}\right]. \tag{23}$$

III. Conservation equation with dissipative source:

$$\dot{\rho}_{\rm SSEE} + 3H\,\rho_{\rm SSEE}(1+w_0) = \frac{9\,\mathrm{KAL}_0\,H^3}{8\pi G}. \tag{24}$$

Derivation of the coefficient 9. From Eq. (11), the viscous pressure is $\Pi = -3\mathrm{KAL}_0 H^2/8\pi G$. In the Raychaudhuri equation the effective pressure of the SSEE sector enters as $p_{\rm eff} = w_0\rho_{\rm SSEE} + \Pi$, so the acceleration term becomes

$$\begin{aligned} -\frac{4\pi G}{3}\big(\rho_{\rm SSEE} + 3p_{\rm eff}\big) &= -\frac{4\pi G}{3}[\rho_{\rm SSEE}(1+3w_0) + 3\Pi] \\ &= -\frac{4\pi G}{3}\left[\rho_{\rm SSEE}(1+3w_0) - \frac{9\,\mathrm{KAL}_0\,H^2}{8\pi G}\right], \end{aligned} \tag{25}$$

which is exactly Eq. (23). The factor 9 arises from $3\times|\Pi|/(H^2) = 3\times 3\mathrm{KAL}_0/8\pi G$; a factor of 3 rather than 9 would correspond to including Π only once instead of as the trace of the viscous stress.

Similarly, the conservation law $\nabla_\mu T^{\mu\nu} = 0$ for the effective fluid (w_0, Π) gives $\dot{\rho} + 3H(\rho + w_0\rho + \Pi) = 0$, hence $\dot{\rho} + 3H\rho(1+w_0) = 9\mathrm{KAL}_0 H^3/8\pi G$, confirming that Eq. (23) and Eq. (24) are mutually consistent.

2.5 Israel–Stewart Viscous Perturbations

The Eckart formulation used in §2.2–2.4 is a first-order theory that violates causality [Hiscock & Lindblom, 1985]: the viscous signal propagates instantaneously. Israel–Stewart (IS) theory [Israel & Stewart, 1979] resolves this by promoting Π to a dynamical variable with a finite relaxation time τ_Π:

$$\tau_\Pi H_0\,\tilde{h}(a)\,\frac{\mathrm{d}\tilde{\Pi}}{\mathrm{d}\ln a} + \tilde{\Pi} = -\,\mathrm{KAL}_0\,\tilde{h}^2(a), \tag{26}$$

where $\tilde{h} = H/H_0$, $\tilde{\Pi} = \Pi\,8\pi G/H_0^2$, and all densities are normalised to $\rho_{\rm crit,0}$. In the limit $\tau_\Pi \to 0$, Eq. (26) reduces to the Eckart result $\tilde{\Pi} = -\mathrm{KAL}_0\tilde{h}^2$.

Causality constraint. The bulk-viscous sound speed in IS theory is [Hiscock & Lindblom, 1985]

$$c_s^2 = \frac{\zeta}{\rho_{\rm DE}\,\tau_\Pi} = \frac{\mathrm{KAL}_0}{3\,\Omega_{\rm DE}\,(\tau_\Pi H_0)} \le 1. \tag{27}$$

Setting $c_s^2 = 1$ (the causality boundary) uniquely fixes the IS relaxation time as an algebraic combination of SSEE constants:

$$\tau_\Pi H_0 \;=\; \frac{\mathrm{KAL}_0}{3\,\Omega_{\rm DE}} \;=\; \frac{\mathrm{KAL}_0\,M_v}{3\,T_r} \;\approx\; 2.191, \tag{28}$$

with no free parameters.

Growth index. Coupling Eq. (26) to the conservation equation (24) and the linear-growth ODE

$$\delta'' + \left[2 + \frac{\mathrm{d}\ln\tilde{h}}{\mathrm{d}\ln a}\right]\delta' = \frac{3}{2}\,\frac{\Omega_{m,\mathrm{dyn}}\,a^{-3}}{\tilde{h}^2(a)}\,\delta, \tag{29}$$

we solve the system numerically (`src/ssee_is_growth.py`) with the IS background $\tilde{h}^2(a) = \Omega_{m,\mathrm{dyn}}\,a^{-3} + \rho_{\mathrm{DE}}(a)$. The growth rate $f \equiv \mathrm{d}\ln\delta/\mathrm{d}\ln a \approx \Omega_m(a)^\gamma$ is well described by a power law over $0.1 \le a \le 1$ with

$$\gamma_{\mathrm{IS}} = 0.657 \pm 0.002 \quad \text{(essentially independent of } \tau_\Pi H_0\text{)}. \tag{30}$$

This differs by 6% from the algebraic sovereignty $\gamma = \varphi^{-1} = 0.618$. Paper 3 §5.4 reports both values: the algebraic prediction $\gamma_{\mathrm{alg}} = 0.618$ and the IS numerical result $\gamma_{\mathrm{IS}} = 0.657$. The IS formalism provides the causal theoretical structure; whether $\gamma = \varphi^{-1}$ is the true attractor of the IS boundary conditions remains a target prediction for forthcoming Stage-IV weak-lensing surveys (LSST, Euclid). Note that the direct IS growth-factor integral (this Appendix) yields $S_8 = 0.837$, while Paper 3's CAMB post-processing method with the same $\gamma_{\mathrm{IS}} = 0.657$ gives $S_8 = 0.818$; both are consistent with Planck 2018 (0.832 ± 0.013) within 1σ.

S_8 constraint. Using the IS growth factor $D_{\mathrm{IS}}(a)$ and the CMB-sector matter density $\Omega_{m,\mathrm{CMB}} = \Omega_{m,\mathrm{dyn}} \times \mathcal{M} = 0.3199$ (MIRA, Paper 3),

$$S_8^{\mathrm{IS}} \equiv \sigma_{8,\mathrm{IS}}\sqrt{\frac{\Omega_{m,\mathrm{CMB}}}{0.3}} \approx 0.837, \tag{31}$$

within 0.4σ of the Planck-2018 value 0.832 ± 0.013 [Planck Collaboration VI, 2020]. Using the dynamic sector $\Omega_{m,\mathrm{dyn}} = 0.160$ instead gives $S_8 \approx 0.59$ (6σ tension); the correct physical choice depends on the formal resolution of the two-sector Friedmann problem (Section 2.8, item 3).

2.6 Physical Regimes

Late universe ($z \lesssim 2$, DESI DR2). At low redshift $H \approx H_0$, so the viscous source $\propto \mathrm{KAL}_0 H^3/8\pi G$ is small compared to ρ_{SSEE}. The acceleration equation is dominated by $\rho_{\mathrm{SSEE}}(1+3w_0) < 0$ (since $w_0 \approx -0.840 < -1/3$), driving $\ddot{a} > 0$ without dark matter. This regime is the one tested against DESI DR2 in Paper 2, where the algebraic prediction (w_0, w_a) matches the data at 0.05σ.

Early universe (CMB epoch, $z \sim 10^3$). At recombination $H \gg H_0$, the geometric retention term $9\,\mathrm{KAL}_0 H^3/8\pi G$ acts as a dissipative source in Eq. (24), injecting energy into the dark-energy fluid and modifying its effective equation of state. This structural viscosity provides the physical mechanism by which an effective matter density $\Omega_{m,\mathrm{eff}} = 1 - T_r/M_v \approx 0.160$ yields the correct CMB acoustic scale through the MIRA factor $\mathcal{M} = 2 - 1/\varphi^2 \approx 1.9989$ (Paper 3, §2.3).

2.7 Inflationary Embedding: α-Attractor with $\alpha = \varphi^4/3$

The SSEE algebraic predictions for the primordial scalar spectral index $n_s = 1 - \varphi^{-7}$ (Paper 1, §1.1) and the tensor-to-scalar ratio $r = \varphi^{-10}$ (Paper 4) belong simultaneously to a well-defined class of inflationary models: the *α-attractors* of Kallosh & Linde [2013].

α-attractor predictions. In the universal leading-order limit, α-attractor models predict

$$n_s = 1 - \frac{2}{N}, \qquad r = \frac{12\alpha}{N^2}, \tag{32}$$

where N is the number of inflationary e-folds and $\alpha > 0$ parameterises the inverse curvature of the scalar field's Kähler target manifold [Kallosh & Linde, 2013, Ferrara et al., 2013]. Setting $\alpha = 1$ recovers Starobinsky R^2 inflation exactly.

SSEE identification. Inserting $n_s = 1 - \varphi^{-7}$ into the first relation gives

$$N = 2/\varphi^{-7} = 2\varphi^7 \approx 58.07, \tag{33}$$

which lies squarely in the observationally required window $50 \lesssim N \lesssim 60$. Substituting $N = 2\varphi^7$ and $r = \varphi^{-10}$ into the second relation:

$$\alpha = \frac{r\,N^2}{12} = \frac{\varphi^{-10} \cdot 4\varphi^{14}}{12} = \frac{4\varphi^4}{12} = \frac{\varphi^4}{3}. \tag{34}$$

This result is algebraically exact (verified numerically to machine precision; see `src/ssee_inflation_co`
Using the Fibonacci identity $\varphi^4 = 3\varphi + 2$:

$$\alpha = \frac{3\varphi + 2}{3} = \varphi + \frac{2}{3} \approx 2.285. \tag{35}$$

Geometric interpretation. In the α-attractor framework the Kähler curvature of the inflaton's target space (a Poincaré disk of radius $\sqrt{3\alpha}$) is $\mathcal{R} = -2/(3\alpha)$. For $\alpha = \varphi^4/3$:

$$\mathcal{R} = -\frac{2}{3 \cdot \varphi^4/3} = -\frac{2}{\varphi^4} = -\varphi^{-4} \approx -0.146. \tag{36}$$

The inflaton therefore lives on a hyperbolic manifold whose curvature is set by φ alone. This gives a *geometric* origin for the appearance of φ in inflationary observables: φ enters not as a numerical coincidence but as the intrinsic length scale of the Kähler geometry of the SSEE inflaton sector.

Slow-roll parameters. The SSEE predictions $(n_s,\, r) = (1-\varphi^{-7},\, \varphi^{-10})$ fix the slow-roll parameters uniquely:

$$\varepsilon = \frac{r}{16} = \frac{\varphi^{-10}}{16}, \tag{37}$$

$$\eta = \frac{n_s - 1 + 6\varepsilon}{2} = \frac{-\varphi^{-7} + 3\varphi^{-10}/8}{2}, \tag{38}$$

$$\frac{\eta}{\varepsilon} = 3 - 8\varphi^3 = -5 - 16\varphi \approx -30.9. \tag{39}$$

The ratio $\eta/\varepsilon = 3 - 8\varphi^3$ is a pure algebraic identity derived from the SSEE axioms $\{\varphi, \pi\}$; it defines a one-parameter family of inflationary potentials $V(\phi_{\rm inf})$ whose single element consistent with α-attractors is $\alpha = \varphi^4/3$.

Relation to Starobinsky. Starobinsky ($\alpha = 1$) reproduces n_s correctly for $N = 2\varphi^7$ but gives $r_{\rm Staro} = 3/\varphi^{14} \approx 0.00356$, a factor of $\varphi^4/3 \approx 2.28$ below the SSEE prediction φ^{-10}. SSEE is therefore *not* Starobinsky inflation; it is a generalised α-attractor that reduces to Starobinsky only in the limit $\varphi \to 1$ (i.e. $\alpha \to 1$, the fixed point of the golden-ratio recursion $x = 1 + 1/x$).

2.8 Summary and Open Problems

Table 2 summarises the EFT status. The action (5) with kinetic exponent (8) and viscosity (14) constitutes a *complete*, zero-free-parameter EFT description of the SSEE background evolution. Every constant (n, KAL_0, w_0, w_a, Ω_{DE}) is derived from φ and π alone; only the overall energy scale H_0 enters as an observational input. Notably, $w_a = -P_{sc}/K_v$ is *not* postulated but emerges as the unique algebraic amplitude of the viscosity's scale-factor evolution (Section 2.3.2).

Four aspects require further development:

1. **Growth index.** The IS-derived growth index $\gamma_{\mathrm{IS}} = 0.657$ (§2.5) differs by 6% from the algebraic prediction $\varphi^{-1} = 0.618$. A mechanism that would select $\gamma = \varphi^{-1}$ from IS boundary conditions is not yet identified; this constitutes a near-prediction testable by LSST/Euclid Stage-IV weak lensing.

2. **Full perturbation spectrum.** The IS growth ODE (29) governs sub-Hubble dark-matter perturbations; the matching to CMB scalar transfer functions and the tensor-mode propagation in the IS background require a complete Boltzmann implementation.

3. **Radiative stability.** The EFT cutoff scale Λ_{UV} and the quantum stability of the $n < 0$ power-law sector are not yet established.

4. **Inflation–dark energy unification.** The α-attractor sector (§2.7) and the k-essence dark-energy sector (§2.2) both originate from φ, suggesting a single unified action. Constructing the quintessential inflation potential $V(\phi)$ that reduces to $P(X) = \mathcal{A}X^n$ at late times and to an $\alpha = \varphi^4/3$ attractor at early times is identified as high-priority future work.

Table 2: EFT status summary for SSEE-V3.6.

Element	Status	Derivation
Action form S	✓Complete	Standard GR + k-essence + Eckart
$u_\mu(\phi)$ coupling	✓Complete	$u_\mu = -\partial_\mu\phi/\sqrt{2X}$
Kinetic exponent n	✓Algebraic	$n = (T_r - M_v)/2T_r$ from φ, π
Equation of state w_0	✓Algebraic	$w_0 = 1/(2n-1) = -T_r/M_v$
Viscosity coupling ζ	✓Algebraic	$\zeta = \mathrm{KAL}_0 H/8\pi G$
Friedmann eqs. I–III	✓Consistent	Factor 9 verified in §2.4
w_a from EFT	✓Algebraic	$\zeta(a)\propto(1-a)\rho_{\mathrm{DE}}/H$; see §2.3.2
IS relaxation time τ_Π	✓Algebraic	$\tau_\Pi H_0 = \mathrm{KAL}_0/(3\Omega_{\mathrm{DE}}) \approx 2.191$
IS causality ($c_s^2 \le 1$)	✓Verified	Eq. (27); $c_s^2 = 1$ at boundary
Growth index γ_{IS}	✓Numerical	0.657 ± 0.002 (6% above $\varphi^{-1} = 0.618$)
S_8 (CMB sector)	✓Numerical	0.837, within 0.4σ of Planck
α-attractor ($\alpha = \varphi^4/3$)	✓Algebraic	Eq. (34); exact, $N = 2\varphi^7$
Kähler curvature $\mathcal{R} = -\varphi^{-4}$	✓Algebraic	Eq. (36)
Full perturbation spectrum	∘ Open	Off-diagonal CMB cov. + Stage-IV WL
Inflation–DE unification	∘ Open	Quintessential inflation potential
Radiative stability	∘ Open	Future work

3 Dark Energy Evolution

SSEE models dynamical dark energy via the Chevallier–Polarski–Linder (CPL) form $w(a) = w_0 + w_a(1 - a)$. Motivated by Eq. (4), the expansion is subject to an effective geometric pressure:

$$H^2 = \frac{8\pi G}{3}(\rho_{\text{bar}} + \rho_{\text{DE}}), \qquad \rho_{\text{DE}} = \rho_{\text{crit},0}\left(\frac{T_r}{M_v}\right). \tag{40}$$

The baseline state w_0 is set by the static geometric ratio T_r/M_v, and the evolution w_a encodes the dynamical departure from saturation driven by P/K_v:

$$w_0 = -\frac{T_r}{M_v} \approx -0.840, \qquad w_a = -\frac{P}{K_v} \approx -0.670. \tag{41}$$

Explicit reduction to the two axioms. Substituting the algebraic definitions $T_r = \frac{3(3\varphi+\pi)}{2}$ (TRIAL), $M_v = 3(\varphi+\pi)$ (MIKAEL_V), $P_{sc} = 2\varphi+\pi$ (PYROS), and $K_v = 2(\varphi+\pi)$ (KRYSTOS), both dark-energy observables collapse to closed forms in φ and π alone:

$$\boxed{w_0 = -\frac{3\varphi + \pi}{2(\varphi + \pi)} = -0.8399}, \qquad \boxed{w_a = -\frac{2\varphi + \pi}{2(\varphi + \pi)} = -0.6699}. \tag{42}$$

No fitting procedure is involved: these values are uniquely fixed by the geometry of φ and π. The numerators $3\varphi + \pi$ and $2\varphi + \pi$ reflect the three- and two-fold contributions of φ to the saturation horizon, while the common denominator $2(\varphi + \pi)$ is twice the Stability Metric $\Omega = \varphi + \pi$.

Asymptotic saturation. Although $w_0 + w_a = -1.510$ would ordinarily imply a phantom Big Rip, SSEE adopts CPL only as a local observational parameterisation. In the far future $(a \to \infty)$ the energy density saturates at the finite T_r/M_v boundary of Eq. (4), ensuring universal stability.

DESI DR2 [Abdul-Karim et al., 2025] analyses indicate an evolving preference for dynamical dark energy. The SSEE algebraic point $(w_0, w_a) = (-0.840, -0.670)$ falls within the 68% confidence contours of the combined BAO+CMB+DESY5 datasets (Mahalanobis $\chi^2_{2D} = 0.080$, i.e. 0.05σ; quantified in Paper 2 [Almeida Vallejo, 2026]).

4 Galaxy Cluster Dynamics

4.1 The Cluster Mass Discrepancy and IGIMF

In rich galaxy clusters, inferred dynamical mass typically exceeds the visible baryonic mass by a factor of ∼5–6. Recent literature incorporating a variable IGIMF [Zhang et al., 2026] demonstrates that baryonic masses have been systematically underestimated: heavy stellar remnants raise the effective baryonic baseline to at least 88^{+5}_{-4}% of the MOND dynamical mass. In SSEE, this updated IGIMF baseline multiplied by $\text{KAL}_0 \approx 5.5214$ natively closes the remaining discrepancy without cold dark matter.

4.2 Structural Viscosity and the Neutrino Bridge

At the cluster scale, the total effective dynamical mass M_{dyn} within radius R is:

$$M_{\text{dyn}}(R) = M^{\text{IGIMF}}_{\text{bar}}(R) \cdot \text{KAL}_0 \cdot [1 + f_\nu], \tag{43}$$

where f_ν quantifies the fractional mass contribution of trapped relic neutrinos. Using $\sum m_\nu = 0.085\,\mathrm{eV}$ — robustly permitted within SSEE's dynamic dark-energy regime — the cosmological neutrino density is $\rho_\nu \approx 28.6\,\mathrm{eV\,cm^{-3}}$. For a cluster with escape velocity $v_{\mathrm{esc}} \approx 2000\,\mathrm{km\,s^{-1}}$, integration of the Fermi–Dirac distribution [Tremaine & Gunn, 1979] yields a trapped fraction $f_\nu \approx 0.018$–0.022. This structurally consistent $\sim$2% correction provides the final macroscopic bridge.

4.3 Quantitative Cluster Analysis

Applying Eq. (43) to four well-documented clusters demonstrates predictive accuracy against observed dynamical masses (Table 3).

Table 3: Cluster mass predictions. All masses within R_{200} in units of $10^{14}\,M_\odot$. $M_{\mathrm{SSEE}} = M_{\mathrm{bar}}^{\mathrm{IGIMF}} \times 5.5214 \times 1.02$. IGIMF baselines and observed masses from Zhang et al. [2026]; Bullet lensing mass from Clowe et al. 2006.

Cluster	$M_{\mathrm{bar}}^{\mathrm{IGIMF}}$ $[10^{14}\,M_\odot]$	M_{SSEE} $[10^{14}\,M_\odot]$	$M_{\mathrm{dyn}}^{\mathrm{obs}}$ $[10^{14}\,M_\odot]$
Coma	1.8 ± 0.2	10.1 ± 1.1	9.8 ± 1.0
A2029	2.2 ± 0.2	12.4 ± 1.1	12.0 ± 1.2
A478	1.5 ± 0.2	8.4 ± 1.1	8.0 ± 1.0
Bullet (main)	1.2 ± 0.2	6.7 ± 1.1	6.5 ± 1.0

All four predictions agree with observed masses at the 1σ level ($\chi_r^2 = 0.122$; full analysis in Paper 2 [Almeida Vallejo, 2026]).

4.4 Gravitational Lensing and the Newtonian Limit

The Bullet Cluster (1E 0657-56) is frequently cited as proof of particulate dark matter due to the spatial offset between collisional X-ray gas and lensing-mass peaks [Clowe et al., 2006]. SSEE reproduces this topological offset through macroscopic structural viscosity without collisionless particles.

The structural viscosity acts as an environment-dependent scalar field. Defining $x = |\nabla\Phi|/a_0$, SSEE adopts an interpolation function analogous to the canonical MOND simple interpolating function $\mu(x)$:

$$\mu_{\mathrm{SSEE}}(x) = \frac{x+1}{x+\mathrm{KAL}_0} \implies \mathrm{KAL}(x) = \frac{1}{\mu_{\mathrm{SSEE}}(x)} = \frac{x+\mathrm{KAL}_0}{x+1}. \tag{44}$$

For $x \gg 1$ (Newtonian limit), $\mathrm{KAL}(x) \to 1$. For $x \ll 1$ (deep MONDian limit), $\mathrm{KAL}(x) \to \mathrm{KAL}_0$. During a high-velocity cluster collision, the stripped X-ray gas flattens its local potential gradient ($x \ll 1$), activating the full KAL_0 amplification; collisionless galaxies retain compact potential wells ($x \geq 1$), coupling non-linearly.

The effective lensing convergence is:

$$\kappa(\vec{\theta}) = \frac{\Sigma_{\mathrm{SSEE}}(\vec{\theta})}{\Sigma_{\mathrm{crit}}}, \quad \Sigma_{\mathrm{SSEE}}(\vec{\theta}) = \int \rho_{\mathrm{bar}}^{\mathrm{IGIMF}}(\vec{\theta},\ell)\cdot \mathrm{KAL}(x)\,d\ell, \quad \Sigma_{\mathrm{crit}} = \frac{c^2}{4\pi G}\frac{D_S}{D_L D_{LS}}. \tag{45}$$

This allows lensing peaks to track galaxies, consistent with QUMOND simulations [Angus et al., 2007].

Data Availability

All observational data used in this work are publicly available. DESI DR2 BAO measurements are from Abdul-Karim et al. [2025] (`https://arxiv.org/abs/2503.14738`). Galaxy-cluster dynamical masses are from Zhang et al. [2026] (`https://arxiv.org/abs/2602.06082`). Analysis scripts and data files are available at `https://github.com/mikealmeida1721/SSEE`.

Acknowledgements

The author thanks the DESI Collaboration for public DR2 data and P. Kroupa and collaborators for the IGIMF cluster re-analysis.

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A Master Derivation Chain: $\varphi, \pi \to$ Observables

Table **??** provides the complete, unambiguous derivation chain from the two axioms (φ, π) to every cosmological observable used in Papers 1–4. This table is the authoritative cross-reference for all four manuscripts. Each intermediate constant is labelled as either a *primary register* (derived in one algebraic step from φ and π) or a *secondary register* (derived from primary registers).

Table 4: Master derivation chain for the
SSEE
framework. Result types: **A** = pure algebraic (no data input), **M** = MCMC-validated (data-constrained posterior), **P** = prior-dependent (requires external dataset anchor). $\varphi = (1+\sqrt{5})/2 \approx 1.61803$; $\pi \approx 3.14159$.

Symbol	Definition	Value	Observable link	Type
Level 0 — Axioms				
φ	$(1+\sqrt{5})/2$	1.618034	All	A
π	Archimedes constant	3.141593	All	A
Level 1 — Primary Registers				
Ω_{DNAV}	$\varphi+\pi$	4.759627	Stability metric	A
β	$(\varphi+\pi)/2$	2.379814	Base coupling	A
P_{sc}	$\Omega_{\mathrm{DNAV}}+\varphi$	6.377661	Dyn. evolution	A
AURA	2φ	3.236068	Resonance scale	A
Level 2 — Secondary Registers				
KAL_0	$\beta+\pi$	5.521406	Viscosity; M_{cluster}	A
K_v	$\varphi+\pi+\Omega_{\mathrm{DNAV}}$	9.519254	Struct. constraint	A
T_r	$3(\varphi+\beta)$	11.993520	Saturation horizon	A
MIRA	$\mathrm{AURA}/2=\varphi$	1.618034	CMB–dyn. bridge	A*
M_v	$\varphi+\pi+K_v$	14.278880	Max. dim. invariant	A
Level 3 — Cosmological Observables				
w_0	$-T_r/M_v$	-0.8399	DESI DR2	A
w_a	$-P_{sc}/K_v$	-0.6699	DESI DR2	A
H_0^{alg}	$3(\varphi+\pi)^2$	67.962 km/s/Mpc	Planck 2018	A
H_0^{MCMC}	emcee posterior	66.75 ± 0.44	DESI+Planck	M
Ω_m^{CMB}	$(\pi-\varphi)/(\pi+\varphi)$	0.3201	Planck 2018	A
n_s	$1-\varphi^{-7}$	0.96556	Planck 2018	A
$\Omega_b h^2$	$3(\pi-\varphi)/200$	0.02285	Planck 2018	A
$\Omega_c h^2$	$\mathrm{KAL}_0 \times \Omega_b h^2 \times n_s$	0.11926	Planck 2018	A
r	φ^{-10}	0.00813	LiteBIRD 2032	A
$\sum m_\nu$	(Acoustic Residue)	0.084 eV	Planck+BBN	A

* MIRA serves a dual role: (i) as a *derived constant* (MIRA $=\varphi$, exact algebraic identity), and (ii) as a *sector-mapping operator* projecting $\Omega_{m,\mathrm{dyn}} = 0.160$ onto $\Omega_{m,\mathrm{CMB}} = 0.3199$ via $\Omega_{m,\mathrm{CMB}} = 2 \times \mathrm{MIRA} \times \Omega_{m,\mathrm{dyn}}$. The dual role is physically meaningful: the same geometric ratio φ governs both the internal structure of the hierarchy and the observational frequency of the CMB sector. Similarly, KAL_0 is both a derived constant (Level 2) and a dynamical viscosity operator scaling cluster masses.

B Unified Symbol Glossary

To avoid ambiguity across Papers 1–4, Table **??** provides the canonical definition of every non-standard symbol used in this series.

Table 5: Unified symbol glossary for the
SSEE
four-paper series. All symbols are dimensionless unless otherwise stated.

Symbol	Full name	First defined
φ	Golden ratio $(1+\sqrt{5})/2$	Paper 1 §1
Ω_{DNAV}	Dark Navigation scalar $(\varphi+\pi)$	Paper 1 §1
β	Base coupling scalar	Paper 1 §1
AURA	Acoustic Unified Resonance Amplitude (2φ)	Paper 1 §2
MIRA	Mapped Inference Resonance Amplitude $(\mathrm{AURA}/2=\varphi)$	Paper 3 §2.4
KAL_0	Asymptotic structural viscosity factor $(\beta+\pi)$	Paper 1 §1
P_{sc}	Dynamical evolution scalar $(\Omega_{\mathrm{DNAV}}+\varphi)$	Paper 1 §1
K_v	Structural constraint invariant	Paper 1 §1
T_r	3D saturation horizon $(3(\varphi+\beta))$	Paper 1 §1
M_v	Maximal dimensional invariant $(\varphi+\pi+K_v)$	Paper 1 §1
PYROS	Radiation sector coupling $(\pi^2/6)$	Paper 4 §3
ÎCËBÊRG	M_v (Nine Sovereignties sum)	Paper 4 §4
$\Omega_{m,\mathrm{dyn}}$	Dynamic matter density (SSEE sector)	Paper 2 §2
$\Omega_{m,\mathrm{CMB}}$	CMB-observed matter density $(2\,\mathrm{MIRA}\,\Omega_{m,\mathrm{dyn}})$	Paper 3 §2.4
γ_{IS}	Israel–Stewart growth index (0.657 ± 0.002)	Paper 1 App.A
$\tau_{\Pi}H_0$	IS relaxation time (dimensionless, ≈ 2.191)	Paper 1 App.A

C Result-Type Classification Protocol

A recurring risk in multi-paper series is that results of different epistemic status are compared without adequate labelling. To ensure methodological transparency across the four manuscripts, all numerical results in this series are classified into one of three categories (see also Table ??, column "Type"):

- **Type A — Algebraic:** The value follows solely from φ and π through the Level 1–3 chain in Table ??. No observational data enters the computation. These results are *prior-free predictions.*
- **Type M — MCMC-validated:** The value is the median of the posterior distribution from the 100-walker `emcee` pipeline described in Paper 2, using DESI DR2 BAO, Planck 2018 priors, and galaxy-cluster mass data. These results reflect the empirical range accessible within the SSEE parameter family.
- **Type P — Prior-dependent:** The value depends on an external dataset anchor (e.g., Planck TT+TE+EE+lensing as a likelihood) that is not internally derived by the SSEE framework. Type P results quantify the consistency of SSEE with standard cosmological datasets; they are not Type A predictions.

The distinction is particularly important for the ΔBIC comparisons in Papers 2 and 3. In Paper 2, the $\Delta\mathrm{BIC}=-5.55$ (dynamic sector) is a Type M result: it reflects the MCMC posterior relative to ΛCDM under identical priors. In Paper 3, the $\Delta\mathrm{BIC}=-40.3$ (plik_lite likelihood) is a Type P result: it depends on the `plik_lite` likelihood as an external CMB anchor. These two BIC values are *not directly comparable* because they operate in different model spaces; the distinction is disclosed explicitly in Paper 3 §4.4.